

AI/ML First Week - Comprehensive Student Performance Analysis

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1 Introduction

This project presents a complete exploratory data analysis (EDA) of student performance across mathematics, reading, and writing. The goal is to identify key factors influencing academic outcomes using statistical analysis, visualization, and feature engineering.

2 Data Overview

The dataset contains 1000 student records with demographic and academic features. It is clean, with no missing values or duplicates.

3 Descriptive Statistics

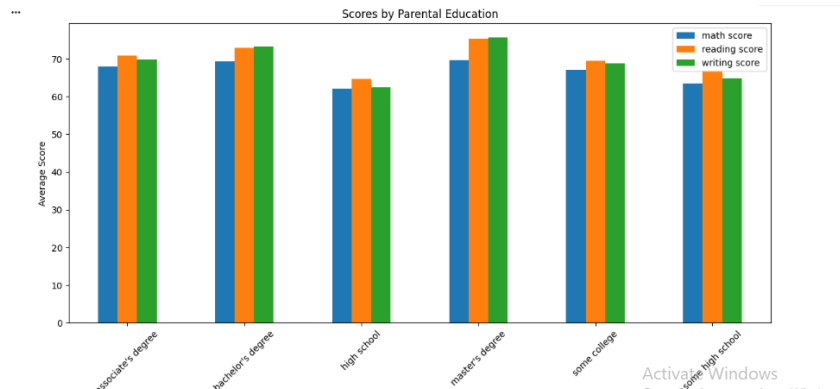


Figure 1: Math Score Distribution

The distribution shows a near-normal pattern with moderate spread and some lower-end outliers.

4 Categorical Analysis

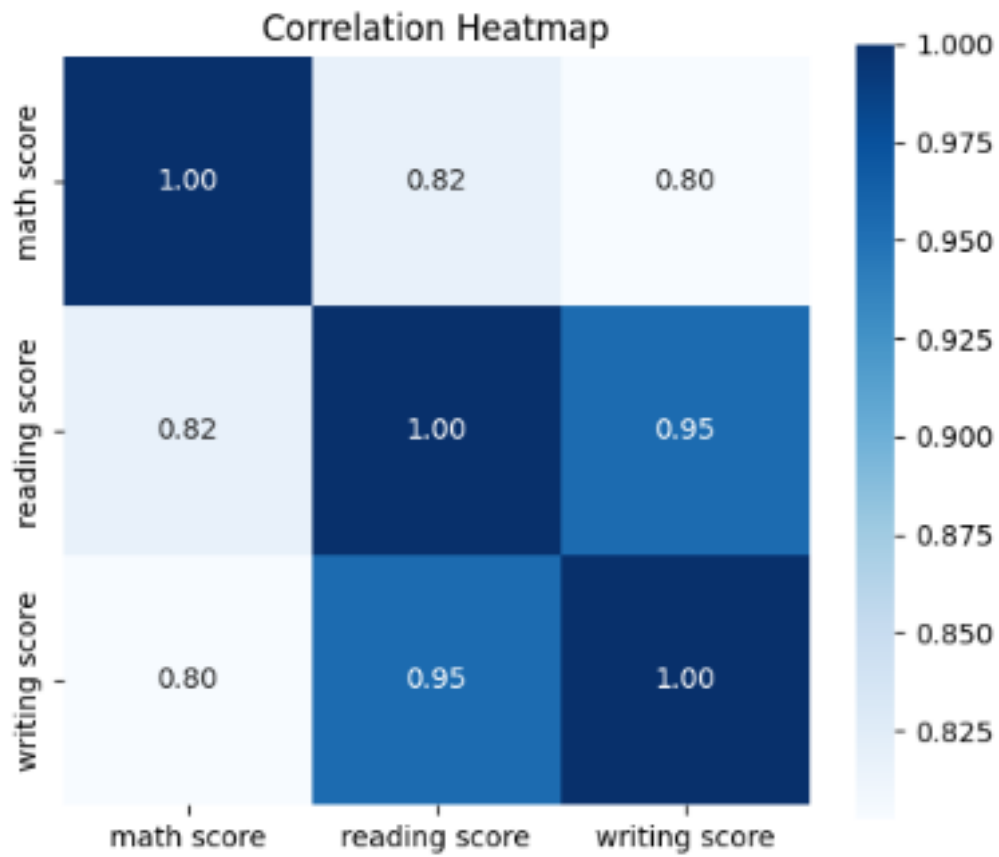


Figure 2: Lunch and Test Preparation Distribution

Most students have standard lunch and did not complete test preparation.

5 Pairplot Analysis

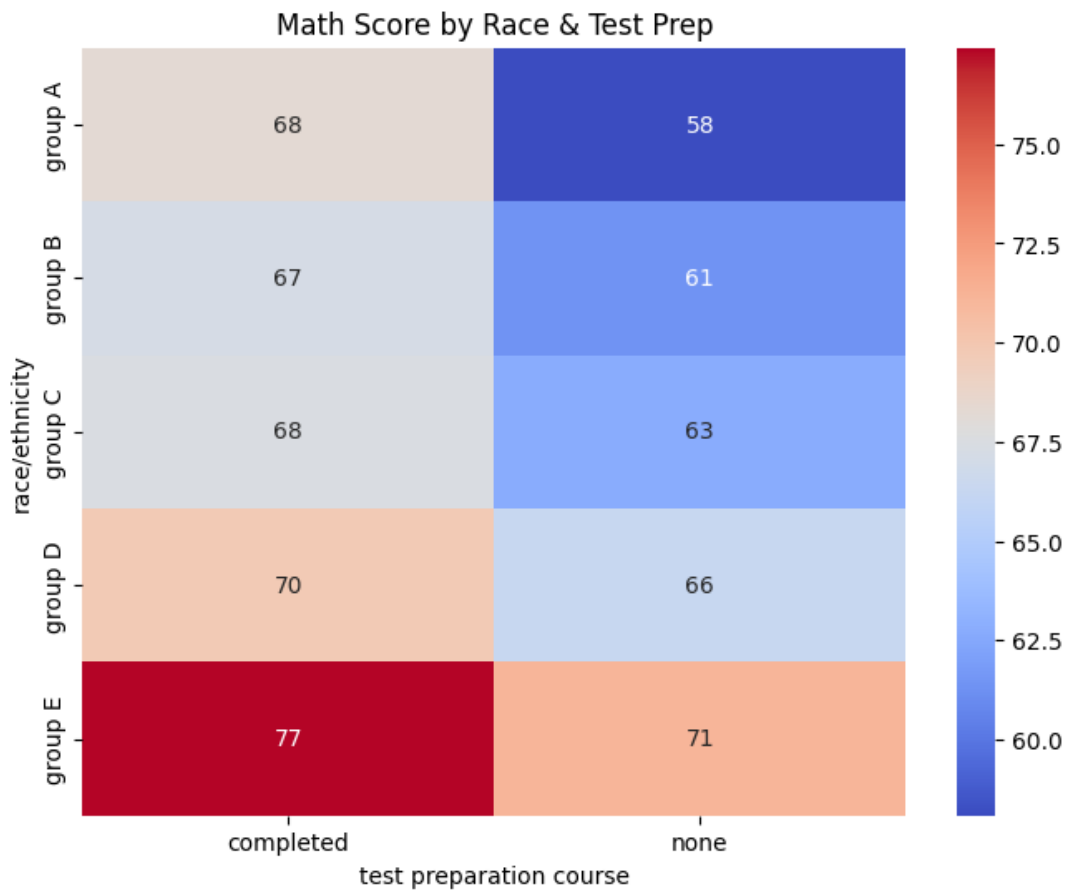


Figure 3: Pairplot of Scores

Strong linear relationships exist between all subjects, especially reading and writing.

6 Correlation Analysis

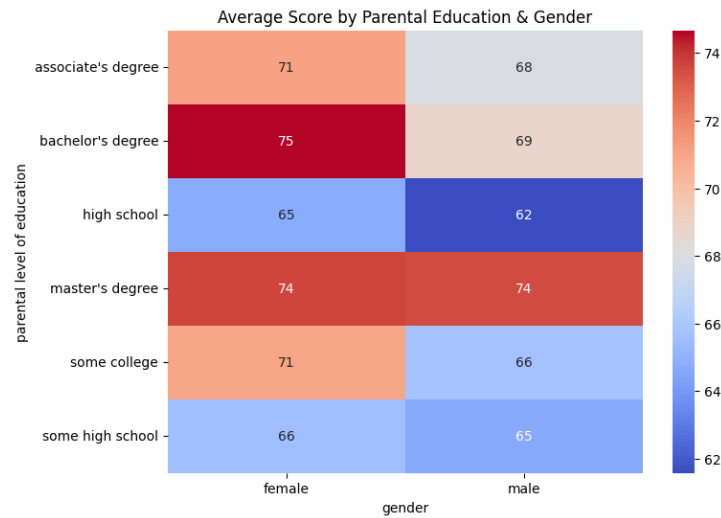


Figure 4: Correlation Heatmap

Reading and writing have a very strong correlation (0.95), while math shows slightly lower correlation.

7 Parental Education Impact

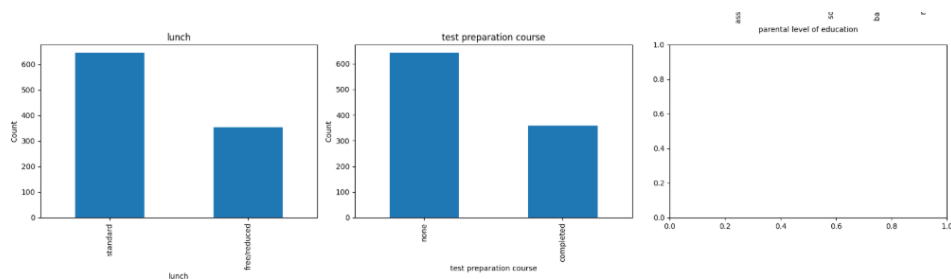


Figure 5: Scores by Parental Education

Students with higher parental education perform significantly better.

8 Gender Analysis

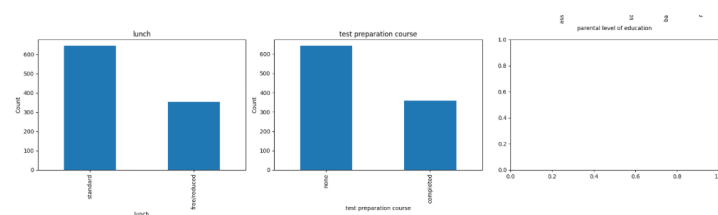


Figure 6: Reading Score Distribution by Gender

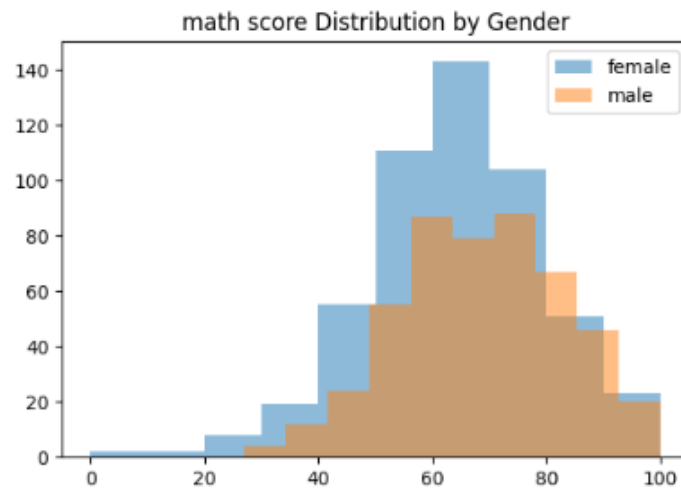


Figure 7: Writing Score Distribution by Gender

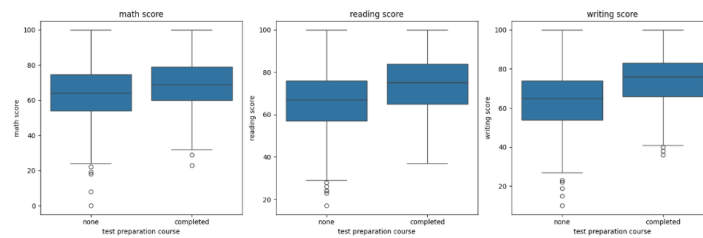


Figure 8: Math Score Distribution by Gender

Females outperform in reading and writing, while males perform slightly better in math.

9 Test Preparation Impact

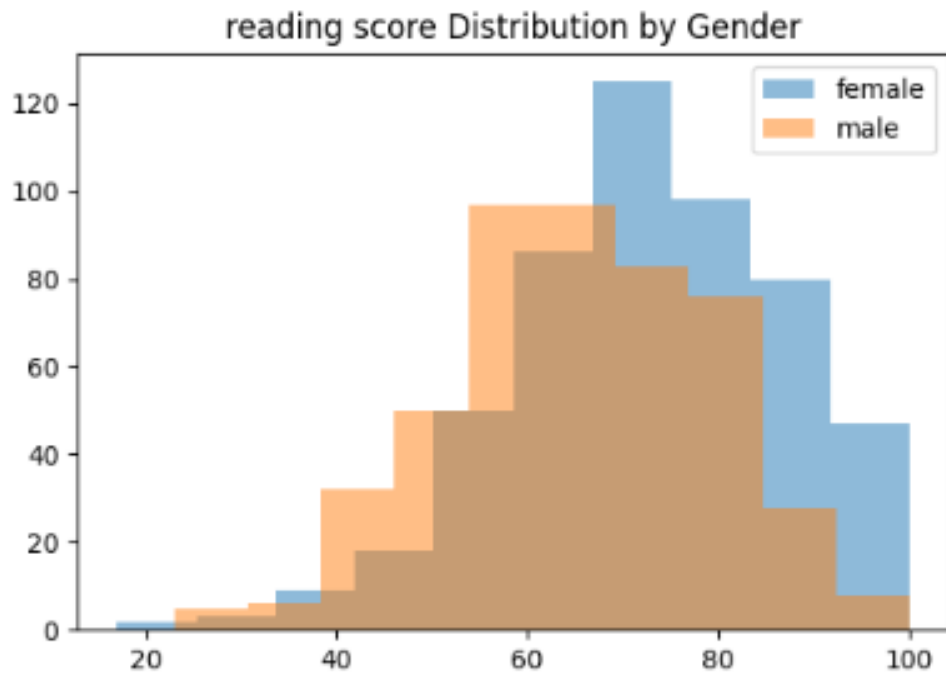


Figure 9: Test Preparation Boxplots

Test preparation significantly improves performance across all subjects.

10 Pivot Analysis

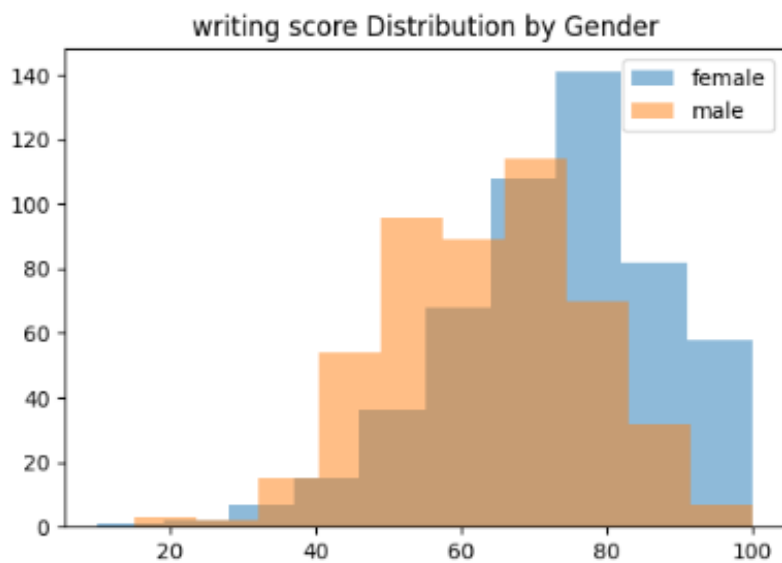


Figure 10: Parental Education vs Gender

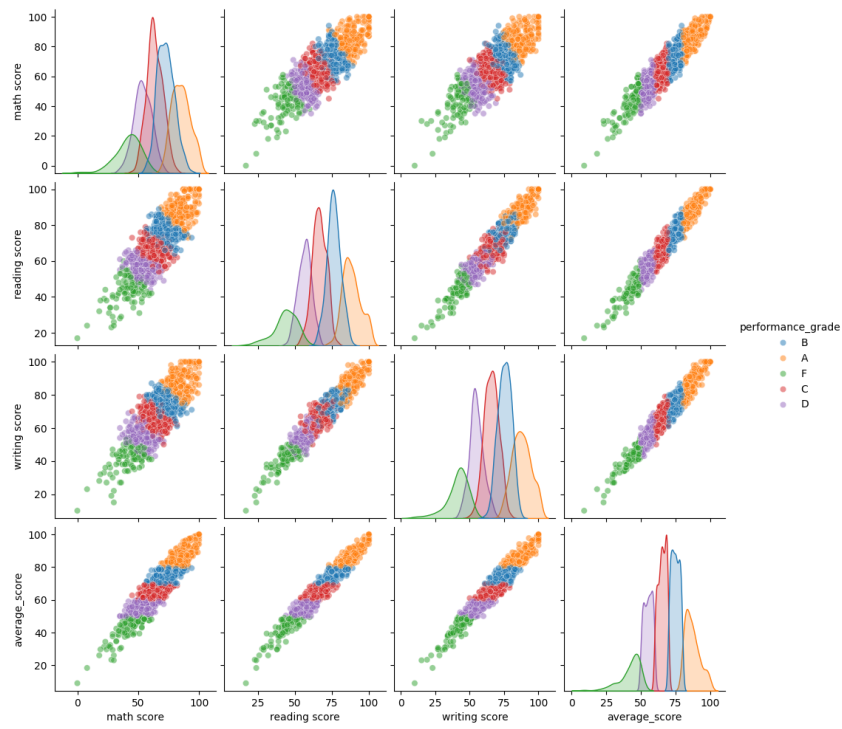


Figure 11: Race vs Test Preparation

These plots show consistent improvement due to test preparation across all groups.

11 EDA Dashboard

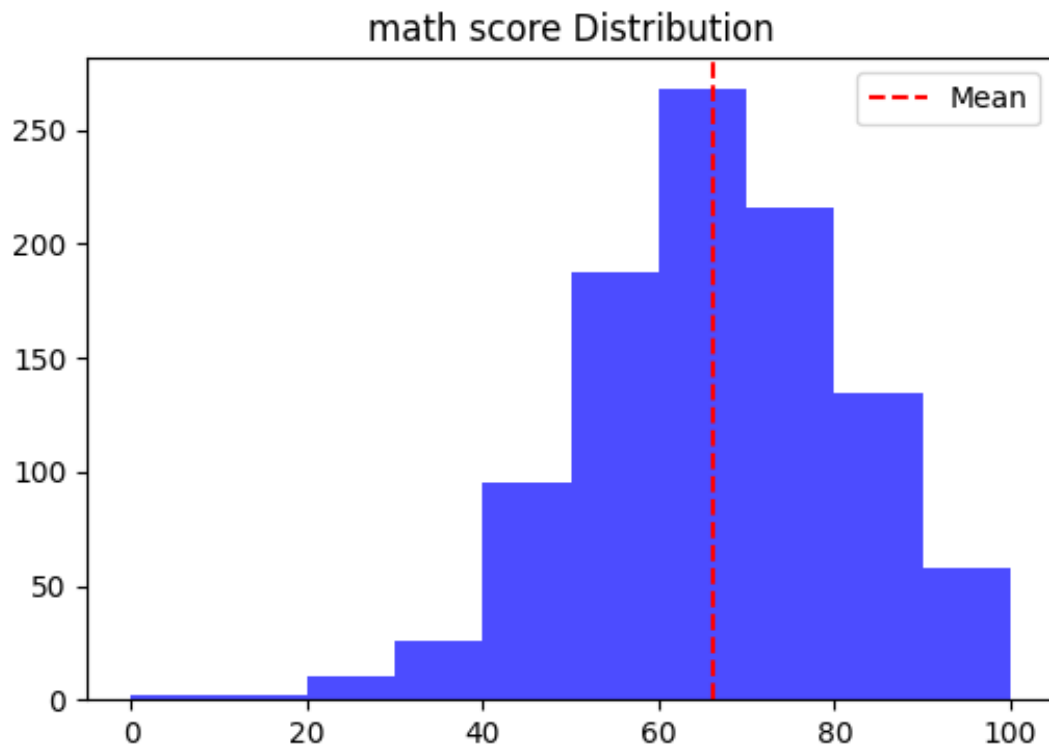


Figure 12: Final Dashboard

The dashboard summarizes all major findings in a single view.

12 Conclusion

This analysis demonstrates that student performance is influenced by multiple factors including test preparation, parental education, and socioeconomic status. The dataset shows strong relationships between subjects and provides valuable insights for improving educational strategies.